

An adaptive fractional order controller design: A realization for liquid level regulation in liquid level plant

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ABSTRACT

In this paper, an adaptive $FO - I^{\lambda}D^{1-\lambda}$ controller design is proposed using a frequency domain approach. The proposed adaptive controller parameters are tuned based on the frequency domain specifications. To realize the velocity error constant for the proposed controller, Oustaloup Recursive Approximation (ORA) method is employed. The proposed controller is realized for liquid level regulation in the Liquid Level Plant (LLP). An identification technique is proposed to identify a linear adaptable parameter transfer function (LAPTF) model of LLP. The performance of the proposed controller is compared with the existing works. It is observed that the proposed controller outperforms the existing controllers in the literature.

1. Introduction

Fractional order calculus (FOC) in engineering field is typically promoted by control engineering applications [1]. In recent years, FOC applications in control engineering have been treated as emerging topics in modeling and controlling of real-time systems. FOC can provide better alignment between the traditional first order and second order integration or differentiation aspects of controllers [2]. Especially, linear fractional order (FO) PID controllers become more popular among all type of FO controllers. In this connection, researchers have been trying to develop FOPID family controllers such as FOPI, modified FOPI, FOPD, modified FOPD and FOPID controllers both in time domain [2,3] and frequency domain [2,4]. FOPID family controllers design became prominent when compared to the conventional PID controllers design [5,6]. However, most of the time domain FOPID family control algorithms are designed based on optimization techniques. Whereas, in frequency domain approaches, the most popular methods used are loop shaping, iso-damping and vector analysis etc which are non-optimization methods. Therefore, FOPID family controllers design using frequency domain approaches are ahead in comparison with time domain approaches [2–4]. Thus, in recent years, researchers give more focus to develop FOPID family controllers algorithms using frequency domain approaches. From the literature works in Refs. [2,7,8], it is observed that all the linear controllers such as conventional and fractional PID family controllers are designed using either time domain

specifications or frequency domain specifications. However, design of the linear controllers using time domain specifications are popularly proposed using optimization methods. Employing optimization methods is highly depends on initial guess of particles. It will be a time taking process to tune the parameters of controllers as the global values of optimization methods can be obtained after many iterations. This time taking process may not support the implementation of linear controllers in adaptive manner. To address the above problem, researchers are focused to develop adaptive control algorithms using frequency domain specifications such as phase margin, gain crossover frequency, gain margin and robustness along with high frequency noise rejection, output disturbance rejection (sensitivity), noise and disturbance rejection and plant sensitivity constraints to tune the FOPID family controllers [2]. However, very few frequency domain specifications works are proposed to tune the control parameters using optimization. In such works, it is observed that constraints are taken as specifications due to which the works forced to approach optimization methods. From the literature works in frequency domain design approaches [2,7,8], it is observed that velocity error constant (K_v) and acceleration error constant (K_a) specifications are not yet used to design adaptive linear fractional order controllers. K_v and K_a specifications play a key in designing the linear controllers. Especially, when the reference signal is time varying [9]. K_v and K_a specifications can be easily employed in conventional integer order PID family controllers design. Whereas, designers have been faced difficulty to employ K_v and K_a specifications in FOPID controllers design

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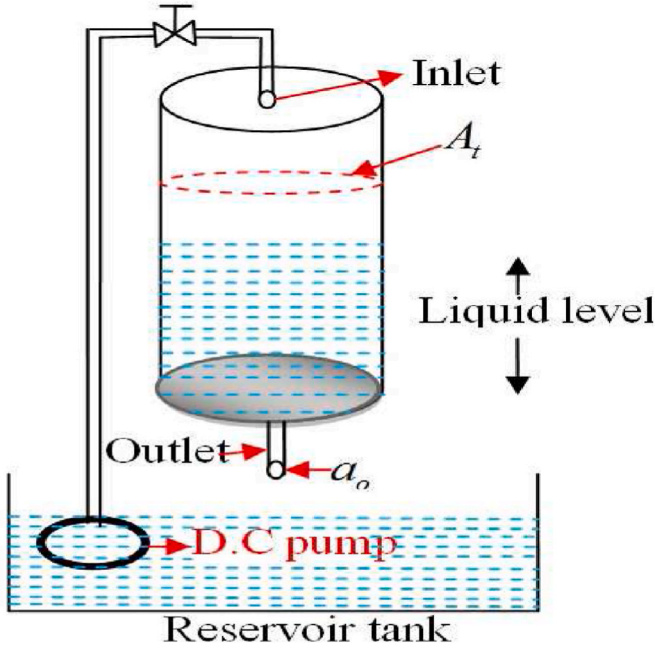


Fig. 1. Schematic diagram of the liquid level system.

[2].

From the above motivations and to address the difficulty of K_v and K_a specifications realization in FOPID family controllers design, an adaptive $FO - I^{\lambda}D^{1-\lambda}$ controller design is proposed using a frequency domain approach. K_v specification is taken as one of the design specifications. To realize K_v specification, Oustaloup Recursive Approximation (ORA) [10] method is employed. As a case study, the proposed controller is implemented for liquid level regulation in Liquid Level Plant (LLP).

2. Linear adaptable parameter transfer function (LAPTF) model identification of LLP

The schematic diagram of LLP is displayed in Fig. 1. It consists of a cylindrical primary tank and a reservoir tank. The DC motor pump in the reservoir tank acts as actuator.

According to Bernoulli's laws, the nonlinear relationship between the output (liquid level in the tank ($h(t)$)) and the input (voltage applied to the DC motor pump ($u(t)$)) of LLP is given as [11]:

$$\frac{dh(t)}{dt} = \frac{-a_o}{A_t} \sqrt{2gh(t)} + \frac{\eta}{A_t} u(t), \quad (1)$$

where a_o is the cross-sectional area of the outlet, A_t is the cross-sectional area of the tank, η is the pump constant which relates the input voltage with the liquid flow from the pump and g is the gravitational constant ($g = 9.81 \text{ m/s}^2$). In the present work, an adaptive $FO - I^{\lambda}D^{1-\lambda}$ controller is proposed to control the liquid level in LLP. Frequency domain specifications are used to tune the parameters of adaptive $FO - I^{\lambda}D^{1-\lambda}$ controller. Hence, a linear transfer function model of LLP is preferable to tune the parameters of the proposed controller using frequency domain specifications [2]. Nonlinear dynamics described in (1) are linearized by using Taylor series expansion. The linearization is performed with respect to h about the steady state operating point of $\bar{h}$. The linearization procedure is as follows:

Let us consider the dynamics described in (1) are further simplified as:

$$\frac{dh(t)}{dt} = -f + nu(t), \quad (2)$$

where

$$n = \frac{\eta}{A_t}, \quad (3)$$

$$f = \left(\frac{a_o}{A_t} \sqrt{2g} \right) \sqrt{h(t)} = k \sqrt{h(t)}. \quad (4)$$

According to Taylor series expansion, if a function $f(x)$ is continuous and differentiable at a point $x = x_0$, then $f(x)$ can be expressed as:

$$f(x) = f(x_0) + \left| \frac{df}{dx_{ss}} \right| \frac{(x - x_0)}{1!} + \left| \frac{d^2f}{dx^2_{ss}} \right| \frac{(x - x_0)^2}{2!} + \dots, \quad (5)$$

where 'ss' refers steady state operating point, i.e., at $x = x_0$. By applying Taylor series expansion and neglecting the higher order terms, (4) is expressed as:

$$f = \bar{f} + \left| \frac{df}{dh_{ss}} \right| (h(t) - \bar{h}). \quad (6)$$

From (4) and (6), one obtains

$$f = \bar{f} + m(h(t) - \bar{h}), \quad (7)$$

where $m = \frac{k}{2}(\bar{h})^{-\frac{1}{2}}$. By substituting (7) in (2), one obtains

$$\frac{dh(t)}{dt} = -\bar{f} - m(h(t) - \bar{h}) + nu(t). \quad (8)$$

At steady state $\frac{dh(t)}{dt} = 0$. Therefore, at the steady state operating point $\bar{h}$, (2) becomes

$$\frac{d\bar{h}}{dt} = -\bar{f} + n\bar{u} = 0. \quad (9)$$

Subtracting (9) from (8), one obtains

$$\frac{d\hat{h}(t)}{dt} = -m\hat{h}(t) + n\hat{u}(t). \quad (10)$$

where $\hat{h}(t)$ and $\hat{u}(t)$ are the deviation variables of $h(t)$ and $u(t)$ respectively and given as:

$$\begin{cases} \hat{h}(t) = h(t) - \bar{h} \\ \hat{u}(t) = u(t) - \bar{u} \end{cases}. \quad (11)$$

Taking Laplace transform of (10), one obtains

$$\frac{\hat{H}(s)}{\hat{U}(s)} = \frac{n}{s + m}. \quad (12)$$

where $\hat{H}(s) = \mathcal{L}\{\hat{h}(t)\}$ and $\hat{U}(s) = \mathcal{L}\{\hat{u}(t)\}$. However, from (12), it is noticed that the parameters m and n are not known exactly. From (1), it is also observed that the un-modelled dynamics such as slothing effect and turbulence effects are not considered in the dynamics of LLP. Therefore, there is a need of adaptable system identification techniques to identify the linear dynamics of LLP. A new identification technique is proposed to identify the linear adaptable parameter transfer function (LAPTF) model of LLP. The LAPTF model of LLP is taken as follows:

$$G(s) = \frac{\hat{H}(s)}{\hat{U}(s)} = \frac{k(\hat{u}(t))}{\tau(\hat{h}(t))s + 1}. \quad (13)$$

The LAPTF model of $G(s)$ is identified from the open-loop real-time response. To identify the LAPTF model, input data set ($\hat{u}(t)$) and output data set ($\hat{h}(t)$) are collected from experimental run of LLP. The parameters $k(\hat{u}(t))$ and $\tau(\hat{h}(t))$ are need to be estimated such that the difference between the response of (13) and real-time response ($\hat{h}(t)$) to be minimized. The minimization problem can be treated as either optimization problem or curve fitting. In the present work, the parameters of k

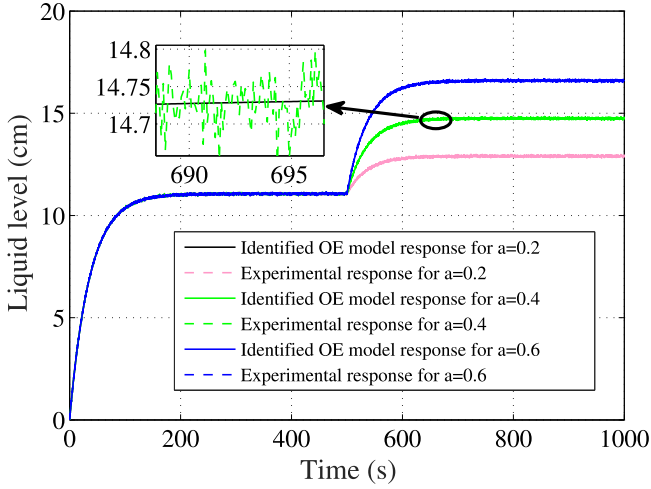


Fig. 2. Experimental and identified LAPTF model response for the input $\hat{u}(t) = 2.85 + a(t - 500)$ volts with $a=0.2, 0.4, 0.6$.

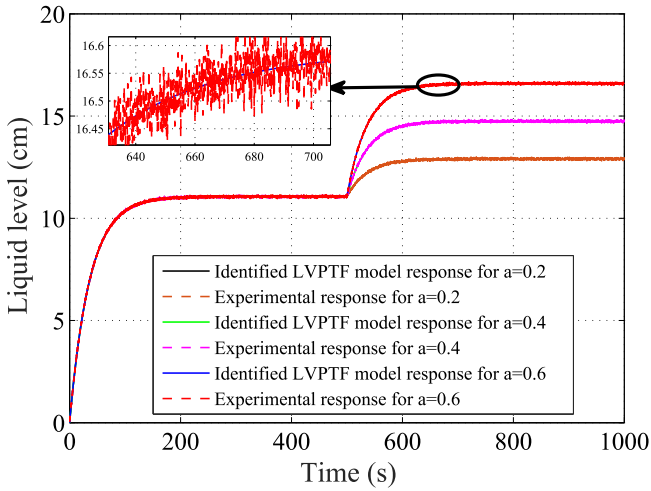


Fig. 3. Experimental and identified OE model response for the input $\hat{u}(t) = 2.85 + a(t - 500)$ volts with $a=0.2, 0.4, 0.6$.

and τ are estimated by using MATLAB curve fitting toolbox and the best fitted k is found as:

$$k(\hat{u}(t)) = -0.002\hat{u}^3(t) + 0.024\hat{u}^2(t) - 0.024\hat{u}(t) + 27.813. \quad (14)$$

The best fitted τ is found as:

$$\tau(\hat{h}(t)) = -0.005\hat{h}^3(t) + 0.029\hat{h}^2(t) - 0.46\hat{h}(t) + 94.82. \quad (15)$$

Fig. 2 shows that the obtained LAPTF model response is almost tracking the real-time experimental response. Therefore, the identified LAPTF model accurately give the dynamics of LLP. In addition, as the LAPTF model has adaptive parameters $k(\hat{u}(t))$ and $\tau(\hat{h}(t))$, therefore, it can allow the designer to design adaptive controller. The proposed identification method is compared with output error (OE) method of identification in Ref. [11]. The identified OE model is obtained as:

$$T(s) = \frac{0.2917}{s + 0.0108} \quad (16)$$

The identified model response of LLP using OE method is given in Fig. 3. A numerical analysis is performed to compare the proposed method with OE method by taking root mean square (RMS) error as performance

index. The RMS error is defined as:

$$E_{rms} = \sqrt{\frac{1}{M} \sum_{j=1}^M (h_r(j) - h_i(j))^2} \quad (17)$$

where h_r is real-time water level, h_i is the water level with identified model, j is the sample number with a sampling interval of 0.1 s and M is the total number of samples. It is found that $E_{rms} = 0.4982$ for proposed LAPTF model and $E_{rms} = 0.9812$ for OE model. It infers that proposed identification method gives better result then OE model.

3. Adaptive $FO - I^{\lambda}D^{1-\lambda}$ controller design

This section presents the proposed controller design methodology. In the present work, an adaptive $FO - I^{\lambda}D^{1-\lambda}$ controller is proposed for liquid level regulation in LLP. The structure of the proposed controller is as follows:

$$C(s) = \frac{K_i + K_d s}{s^{\lambda}}, \quad (18)$$

To regulate the liquid level in LLP, the proposed $FO - I^{\lambda}D^{1-\lambda}$ is suitable than the integer order PID controller due to the following reasons:

- From (18), for $\lambda=1$, the proposed controller acts as PI controller and for $\lambda=0$, the proposed controller acts as PD controller. Therefore, the proposed $FO - I^{\lambda}D^{1-\lambda}$ controller is a trade-off controller between PI and PD controllers.
- From (18), when λ increases, the gain at low operating frequencies also increases. Higher gain at low operating frequency provides good tracking performance [12]. Hence, the proposed $FO - I^{\lambda}D^{1-\lambda}$ controller provides good tracking performance than the PID controller.
- From Fig. 2, it is noticed that the open-loop response of the LLP is non-oscillatory. Therefore, the impact of Derivative-controller is not much needed in non-oscillatory system like LLP [12]. The proposed $FO - I^{\lambda}D^{1-\lambda}$ controller in (18) can also be expressed as:

$$C(s) = K_i s^{-\lambda} + K_d s^{1-\lambda}. \quad (19)$$

If $\lambda \in (1, 2)$ then $1 - \lambda < 0$. Therefore, from (19), it can be observed that the proposed controller provides integral action for $\lambda \in (1, 2)$. Hence, while designing the controller, if we manage the fractional order λ of the $FO - I^{\lambda}D^{1-\lambda}$ controller in the range of $\lambda \in (1, 2)$ then the proposed $FO - I^{\lambda}D^{1-\lambda}$ controller will have integral action.

The frequency domain representation of (18) is as follows:

$$C(j\omega) = \frac{K_i + K_d(j\omega)}{(j\omega)^{\lambda}}. \quad (20)$$

From (20), one obtains

$$C(j\omega) = \frac{1}{\omega^{\lambda}} \left[\cos \frac{\lambda\pi}{2} - j \sin \frac{\lambda\pi}{2} \right] (K_i + jK_d\omega). \quad (21)$$

From (21), the phase of the controller is obtained as

$$\varphi_c = -\frac{\lambda\pi}{2} + \tan^{-1} \left(\frac{K_d\omega}{K_i} \right). \quad (22)$$

From (20), the magnitude of the controller is obtained as

$$|C(j\omega)| = \frac{1}{\omega^{\lambda}} \sqrt{K_i^2 + (K_d\omega)^2}. \quad (23)$$

To tune the parameters of the proposed adaptive $FO - I^{\lambda}D^{1-\lambda}$ controller, a classical frequency domain approach is used. The proposed controller in (18) has three parameters K_i , K_d and λ to be tuned. In general, to tune a

three parameter linear controller, there is a need of obtain three mathematical equations to design the three parameters of the controller [12]. To obtain the three mathematical equations, three specifications such as phase margin (φ_m)= 80° , gain crossover frequency (ω_{gc})= 0.5 rad/s and velocity error constant (K_v)= $1s^{-1}$ are considered as desired specifications.

3.1. Velocity error constant specification significance

In the present work, velocity error constant (K_v) is taken as one of the design specifications to design the proposed controller. From the literature study in Ref. [2], it is observed that FOPID family controllers were popularly designed by considering specifications such as phase margin, gain crossover frequency, gain margin, phase crossover frequency, robustness to gain variations and other constraints. But, in literature works in Ref. [2], K_v specification is not considered which plays an important role when the reference signal is time varying signal. In fact, as λ is integral order of FOPID family controllers, it plays an important role in improving the steady state performance of the closed loop system [9,12]. Particularly from (41), it is going to be observed that integral order of proposed controller λ changes with K_v . Therefore K_v specification plays important role to improve the steady state tracking performance.

Tuning procedure of the proposed controller parameters is as follows:

According to the phase margin (φ_m) specification, the following criterion is defined.

$$\varphi_m = 180^\circ + \varphi \quad (24)$$

where φ is the phase of open loop system. From (24), it will be written as

$$\varphi = -180^\circ + \varphi_m \quad (25)$$

As the proposed controller and the plant is present in the open loop system, (25) will be written as

$$|\varphi_C \omega=\omega_{gc} + \varphi_G \omega=\omega_{gc} = -180^\circ + \varphi_m, \quad (26)$$

where φ_G is the phase of the plant in (13). φ_C is the phase of the controller in (18). From (13), (22) and (26), it will be written as

$$-\frac{\lambda\pi}{2} + \tan^{-1}\left(\frac{K_d\omega_{gc}}{K_i}\right) + \tan^{-1}\{\tau[\hat{h}(t)]\omega_{gc}\} = -180^\circ + \varphi_m \quad (27)$$

From (27), it will be obtained as

$$-\frac{\lambda\pi}{2} + \tan^{-1}\left(\frac{K_d\omega_{gc}}{K_i}\right) = -180^\circ + \varphi_m - \tan^{-1}\{\tau[\hat{h}(t)]\omega_{gc}\} \quad (28)$$

By considering the known value $-180^\circ + \varphi_m - \tan^{-1}\{\tau[\hat{h}(t)]\omega_{gc}\} = \theta$, (28) will be written as

$$-\frac{\lambda\pi}{2} + \tan^{-1}\left(\frac{K_d\omega_{gc}}{K_i}\right) = \theta \quad (29)$$

From (29), it will be obtained as

$$K_d\omega_{gc} = K_i \times \tan\left(\theta + \frac{\lambda\pi}{2}\right) \quad (30)$$

According to the gain crossover frequency (ω_{gc}) specification, the following criterion is defined.

$$|G(j\omega)C(j\omega)|_{\omega=\omega_{gc}} = 1. \quad (31)$$

From (13), (23) and (31), it can be derived as

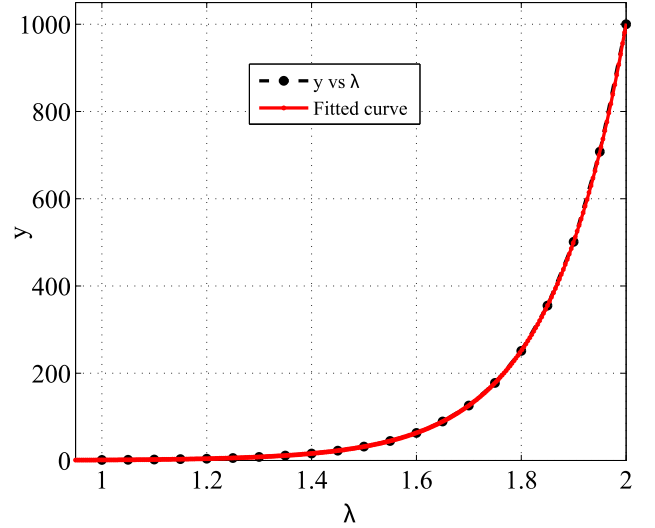


Fig. 4. $y = \text{ORA value of } \lim_{s \rightarrow 0} \frac{1}{s^{\lambda-1}}$ and MATLAB fitted curve vs λ .

$$\frac{k[\hat{u}(t)]\sqrt{K_i^2 + (K_d\omega_{gc})^2}}{\omega_{gc}^{\lambda}\sqrt{\{\tau[\hat{h}(t)]\omega_{gc}\}^2 + 1}} = 1. \quad (32)$$

By substituting (30) in (32), one obtains

$$K_i = \left\{ \frac{\omega_{gc}^{\lambda}\sqrt{\{\tau[\hat{h}(t)]\omega_{gc}\}^2 + 1}}{k[\hat{u}(t)]} \right\} \cos\left(\theta + \frac{\lambda\pi}{2}\right). \quad (33)$$

According to the velocity error constant (K_v) specification, the following criterion is defined.

$$\lim_{s \rightarrow 0} sG(s)C(s) = K_v. \quad (34)$$

By substituting (13) and (18) in (34), it will be written as

$$\lim_{s \rightarrow 0} s \times \frac{k(\hat{u}(t))}{\tau(\hat{h}(t))s + 1} \times \frac{K_i + K_d s}{s^{\lambda}} = K_v \quad (35)$$

From (35), it will be written as

$$k(\hat{u}(t)) \times \lim_{s \rightarrow 0} s \left[\frac{K_i + K_d s}{s^{\lambda}} \right] = K_v \quad (36)$$

From (36), it will be written as

$$k(\hat{u}(t)) \times \lim_{s \rightarrow 0} \left[\frac{K_i + K_d s}{s^{\lambda-1}} \right] = K_v \quad (37)$$

From (37), it will be written as

$$k[\hat{u}(t)] \times K_i \times \lim_{s \rightarrow 0} \frac{1}{s^{\lambda-1}} = K_v. \quad (38)$$

By substituting (33) in (38), one obtains

$$\left\{ \frac{\omega_{gc}^{\lambda}\sqrt{\{\tau[\hat{h}(t)]\omega_{gc}\}^2 + 1}}{k[\hat{u}(t)]} \right\} \times \cos\left(\theta + \frac{\lambda\pi}{2}\right) \times \lim_{s \rightarrow 0} \frac{1}{s^{\lambda-1}} = K_v. \quad (39)$$

The term $\lim_{s \rightarrow 0} \frac{1}{s^{\lambda-1}}$ in (39) is approximated to a finite value by using Oustaloup Recursive Approximation (ORA) [10] method. ORA method and MATLAB curve fitting tool is used to find the finite value of $\lim_{s \rightarrow 0} \frac{1}{s^{\lambda-1}}$. To perform the ORA, the frequency band $[\omega_l \ \omega_h] = [0.001 \ 10]$ rad/s and the order $N = 5$ are chosen as per the rules given in Ref. [13]. The

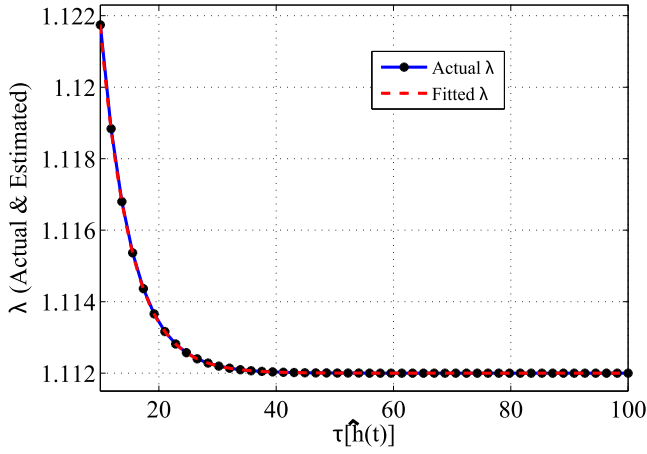


Fig. 5. Curves obtained from the relations given in (41) and (42).

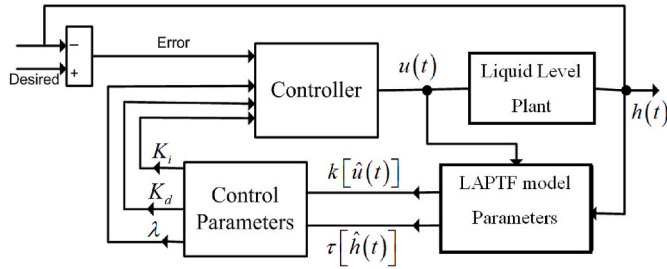


Fig. 6. Schematics diagram of control scheme.

ORA value of $\lim_{s \rightarrow 0} \frac{1}{s^{\lambda-1}}$ is estimated as a function of λ using MATLAB curve fitting.

From the exponentially fitted curve as shown in Fig. 4, it is obtained as:

$$\lim_{s \rightarrow 0} \frac{1}{s^{\lambda-1}} \approx 0.001065e^{6.874\lambda}. \quad (40)$$

By substituting (40) in (39), one obtains

$$\left\{ \omega_{gc}^2 \sqrt{\{\tau[\hat{h}(t)]\omega_{gc}\}^2 + 1} \right\} \times \cos\left(\theta + \frac{\lambda\pi}{2}\right) \times 0.001065e^{6.874\lambda} = K_v, \quad (41)$$

From (41), it is observed that the values of $\tau[\hat{h}(t)]$, θ , ω_{gc} , and K_v are known from the identified model ((13)–(15)) and the chosen specifications. Therefore, (41) can be solved by using 'fsolver' in the MATLAB to find the value of λ . However, equation (41) online solution with 'fsolver' causes delay and computational burden in the process of controller design. It is undesirable. To overcome the problem, the relation between λ and $\tau[\hat{h}(t)]$, given in (41) is approximated by a function $\lambda\{\tau[\hat{h}(t)]\}$ with in the possible range of $\tau[\hat{h}(t)]$. The approximation is performed using MATLAB curve fitting method and $\lambda\{\tau[\hat{h}(t)]\}$ is found as:

$$\lambda\{\tau[\hat{h}(t)]\} = 0.067e^{-0.1928 \times \tau[\hat{h}(t)]} + 1.112. \quad (42)$$

From Fig. 5, it is observed that the curves of λ vs $\tau[\hat{h}(t)]$ by means of the relation given in (41) and (42) are accurately fitted. The estimation function in (42) gives the online value of λ . Once λ is obtained, remaining K_i and K_d parameters are computed using (33) and (30) respectively. Online updation of K_i , K_d and λ is performed in a programmable environment of MATLAB software. The proposed control

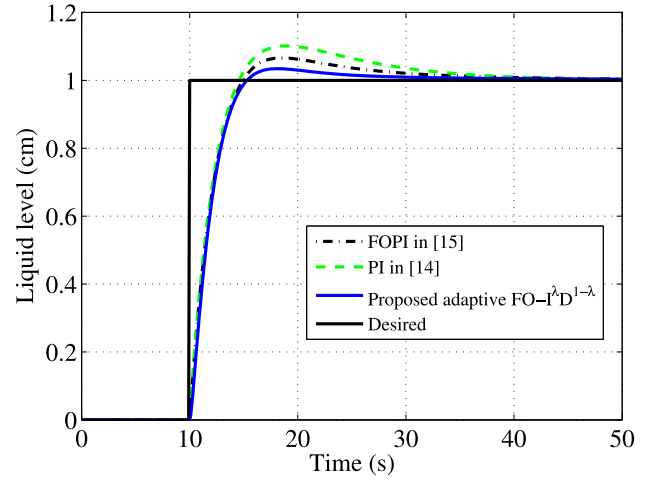


Fig. 7. Closed-loop tracking curves in simulation.

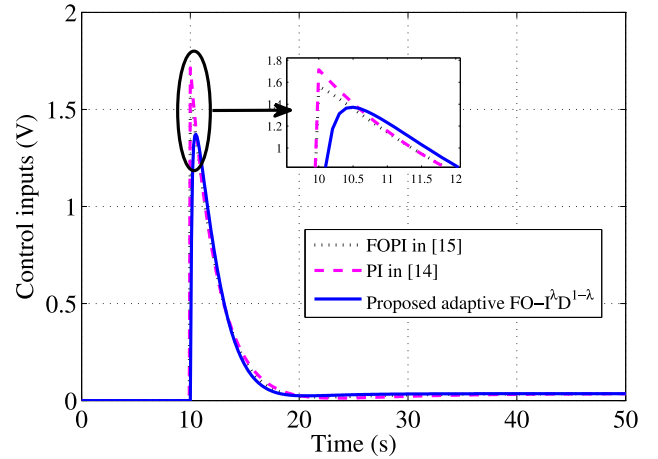


Fig. 8. Control inputs in simulation.

Table 1
Performance analysis in simulation.

Controller	Tracking performance				Control action performance	
	t_s	MP (%)	IAE (cm)	ITAE (cm-s)	$\ u\ _2$ (V)	$ u _{max}$ (V)
FOPI [15]	20.38	6.56	2.51	40.12	6.03	1.56
PI [14]	24.58	10.15	2.93	50.69	6.21	1.71
Proposed adaptive $FO - I^\lambda D^{1-\lambda}$	13.10	3.18	2.11	30.91	5.80	1.37

scheme is shown in Fig. 6.

4. Results and discussions

The proposed adaptive $FO - I^\lambda D^{1-\lambda}$ controller performance is compared with PI controller in Ref. [14] and FOPI controller in Ref. [15]. The performance of the controllers is evaluated both in simulation and real-time. Performance evaluation is performed in the aspects of (i) Tracking performance and (ii) Control action performance. Simulated closed-loop tracking curves of adaptive $FO - I^\lambda D^{1-\lambda}$, PI [14] and FOPI [15] controllers are shown Fig. 7. Simulated control inputs of adaptive $FO - I^\lambda D^{1-\lambda}$, PI [14] and FOPI [15] controllers are shown in

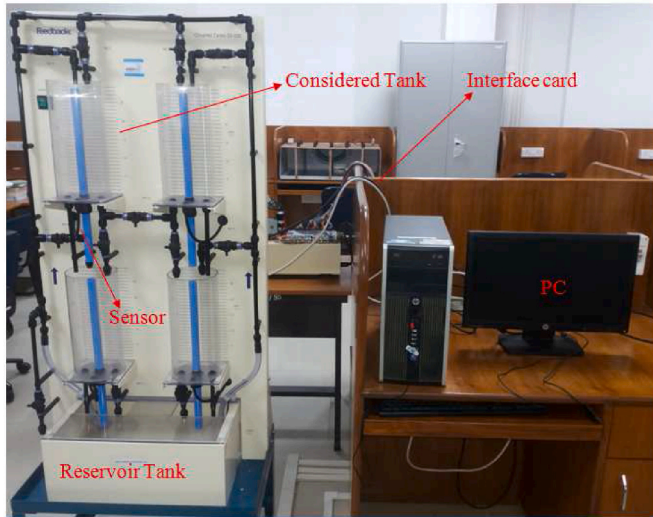


Fig. 9. Real-time LLP.

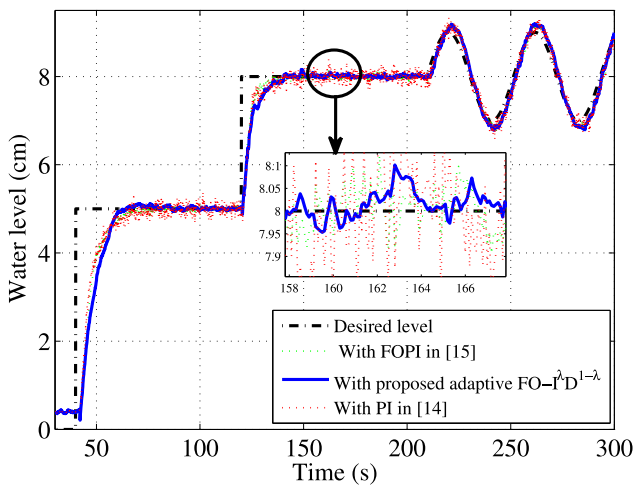


Fig. 10. Real-time closed-loop tracking curves.

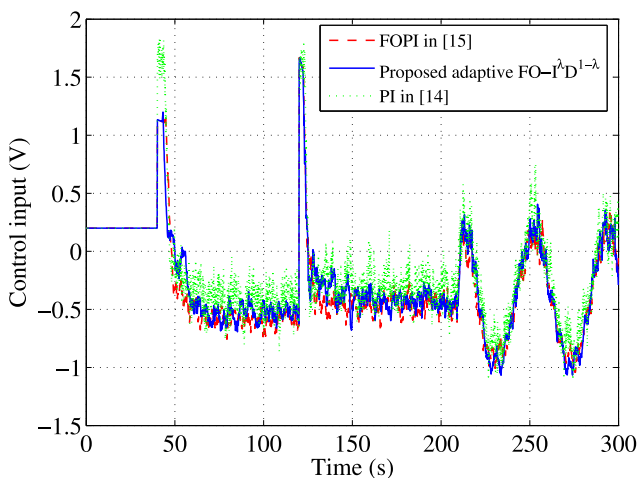


Fig. 11. Real-time control inputs.

Table 2

Performance analysis in real-time.

Controller	IAE (cm)	ITAE (cm-s)	$\ u\ _2$ (V)
FOPI [15]	76.34	875.42	33.56
PI [14]	84.61	960.67	40.43
Proposed adaptive $FO - I^\lambda D^{1-\lambda}$	65.55	545.15	23.01

Fig. 8. Simulated tracking and control input curves numerical analysis is given in Table 1. In Table 1, t_s is settling time with $\pm 2\%$ tolerance band, MP(%) is maximum peak, IAE is Integral Absolute Error $= \int_0^t |e(r)| dt$ and ITAE is Integral Time Absolute Error $= \int_0^t t \times |e(r)| dt$, $\|u\|_2$ denotes the 2-Norm of the control signal and $|u|_{max}$ denotes maximum absolute value of control signal. If $|u|_{max}$ is very high then there might be chances of actuator saturation. If it is low then the actuator saturation chances are less. From Ref. [12], energy required for controller's control action is approximately direct proportional to the square of the 2-norm of control input (u). Therefore, to know the energy efficiency of the proposed controller, $\|u\|_2$ is considered as one of the performance indices. In the aspect of real-time, controllers are implemented for liquid level regulation in real-time LLP as shown in Fig. 9. Real-time closed loop tracking curves are shown in Fig. 10. Real-time control inputs are shown in Fig. 11. The real-time analysis is given in Table 2. From the performance analysis given in Tables 1 and 2 and it is observed that the proposed adaptive $FO - I^\lambda D^{1-\lambda}$ controller gives better performance than the PI [14] and FOPI [15]. From Tables 1 and it can be observed that the $|u|_{max}$ is low for the proposed controller. Therefore the chances of actuator saturation are low with the proposed controller.

5. Conclusion

In this paper, an adaptive $FO - I^\lambda D^{1-\lambda}$ controller design is proposed using a classical frequency domain approach. A mathematical description is presented in detail to tune the proposed controller parameters based on the frequency domain specifications. Oustaloup Recursive Approximation (ORA) method and MATLAB curve fitting tool is employed to realize the velocity error constant for the proposed controller. The proposed controller is implemented for liquid level regulation in the Liquid Level Plant (LLP), both in simulation and real-time. An identification technique is proposed to identify the linear adaptable parameter transfer function (LAPTF) model of LLP. The performance of the proposed controller is compared with the reported works in the literature and it gives better performance than the PI [14] and FOPI [15] controllers. As a future extension of the present work, one may implement the present methodology of controller design for Multiple input multiple output (MIMO) system by employing the decouplers. In way forward, one may work on fractional order variable transfer function modeling of LLP.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] A. Tepljakov, B.B. Alagoz, C. Yerglu, E.A. Gonzalez, S.H. Hosseinnia, E. Petlenkov, A. Ates, M. Cech, Towards industrialization of fopid controllers: a survey on milestones of fractional-order control and pathways for future developments, *IEEE Access* 9 (2021) 21, 016–21 042.
- [2] A.A. Dastjerdi, B.M. Vinagre, Y. Chen, S.H. Hosseinia, Linear fractional order controllers; a survey in the frequency domain, *Annu. Rev. Control* (2019).

- [3] M.S. Ayas, E. Sahin, Fopid controller with fractional filter for an automatic voltage regulator, *Comput. Electr. Eng.* 90 (2021), 106895.
- [4] A.A. Dastjerdi, N. Saikumar, S.H. HosseinNia, Tuning guidelines for fractional order pid controllers: rules of thumb, *Mechatronics* 56 (2018) 26–36.
- [5] T. Pidikiti, B. Gireesha, M. Subbarao, V.M. Krishna, et al., Design and control of takagi-sugeno-kang fuzzy based inverter for power quality improvement in grid-tied pv systems, *Measurement: Sensors* 25 (2023), 100638.
- [6] H. Taghavifar, EKF estimation based pid type-2 fuzzy control of electric cars, *Measurement* 173 (2021), 108557.
- [7] M.E. Krijnen, R.A. van Ostayen, H. HosseinNia, The application of fractional order control for an air-based contactless actuation system, *ISA Trans.* 82 (2018) 172–183.
- [8] A. Chevalier, C. Francis, C. Copot, C.M. Ionescu, R. De Keyser, Fractional-order pid design: towards transition from state-of-art to state-of-use, *ISA Trans.* 84 (2019) 178–186.
- [9] N.S. Nise, *Control Systems Engineering*. Plus 0.5em Minus 0, 4emJohn Wiley & Sons, 2020.
- [10] A. Oustaloup, F. Levron, B. Mathieu, F.M. Nanot, Frequency-band complex noninteger differentiator: characterization and synthesis, *IEEE Trans. Circ. Syst. I: Fundament. Theor. Appl.* 47 (1) (2000) 25–39.
- [11] A.K. Tangirala, *Principles of System Identification: Theory and Practice*. Plus 0.5em Minus 0, 4emCrc Press, 2018.
- [12] P.N. Paraskevopoulos, *Modern Control Engineering*. Plus 0.5em Minus 0, 4emCRC Press, 2017.
- [13] F. Merrikh-Bayat, Rules for selecting the parameters of oustaloup recursive approximation for the simulation of linear feedback systems containing pi/d_μ controller, *Commun. Nonlinear Sci. Numer. Simul.* 17 (4) (2012) 1852–1861.
- [14] S.R. Mahapatro, B. Subudhi, S. Ghosh, Design and experimental realization of a robust decentralized pi controller for a coupled tank system, *ISA Trans.* 89 (2019) 158–168.
- [15] C.I. Muresan, E.H. Dulf, R. Both, Vector-based tuning and experimental validation of fractional-order pi/pd controllers, *Nonlinear Dynam.* 84 (1) (2016) 179–188.